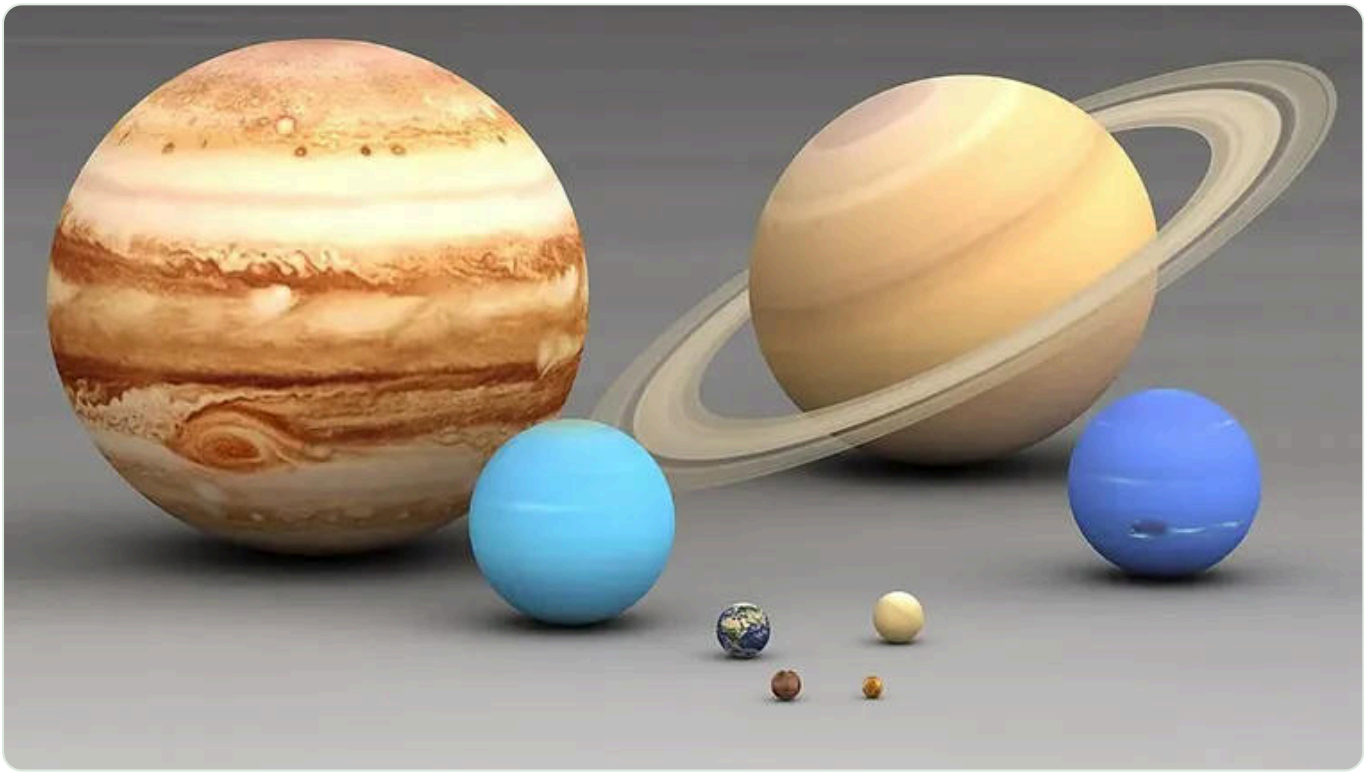
Sindh Science Learning Portal

Class 5 · Science

Chapter 10

Earth and Space



Learn about the solar system, how day and night happen, what a year is, and why the moon seems to change shape.

Following the Sindh Textbook Board curriculum

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CHAPTER 10

Earth and Space

What we will learn Learn about the solar system, how day and night happen, what a year is, and why the moon seems to change shape.

1 The solar system

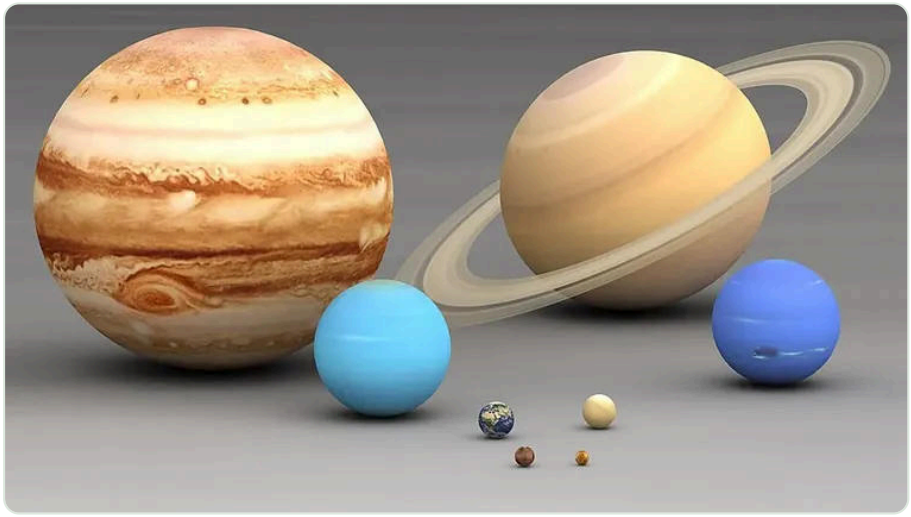
The **sun** is a **star**. It is a huge ball of hot gases that gives out light and heat.

The sun and the **planets**, moons and other objects that go around it make up the **solar system**.

Starting from the sun, the eight planets are: **Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune**.

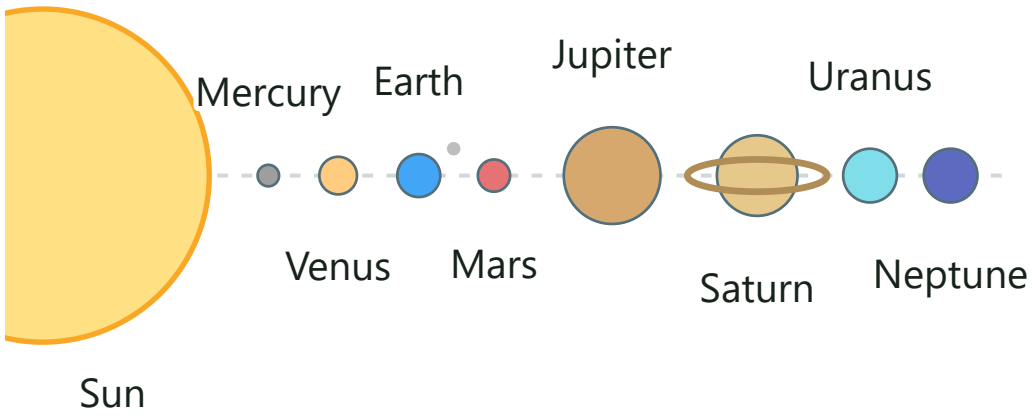
Our **Earth** is the only planet where we know there is life, because it has water, air and the right amount of heat.

Jupiter is the biggest planet and Mercury is the closest to the sun.



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Diagram: the parts and what they do



Sun (سج · سورج)
A star that gives light and heat to the solar system.

Mercury (عطارد · عطارد)
The closest planet to the sun and the smallest.

Venus (زهره · زہرہ)
The hottest planet. It shines like a bright star in the morning or evening.

Earth (زمين · زمین)
Our home: the planet with water, air and life. It has one moon.

Mars (مريخ · مریخ)
The red planet, because its soil has rust in it.

Jupiter (مشتري · مشتری)
The biggest planet, made mostly of gases.

Saturn (زحل · زحل)
It has beautiful rings made of ice and rock.

Uranus (يورينس · یورینس)
A very cold planet that spins lying on its side.

Neptune (نيپچون · نیپچون)
The farthest planet from the sun, with very strong winds.

2 Day, night and year

The Earth spins like a top on its **axis** (an imaginary straight line). This is called **rotation**. One full spin takes about **24 hours**.

The side of the Earth facing the sun has **day**, and the side facing away from the sun has **night**.

The Earth also travels around the sun. One full trip around the sun takes about **365 days** — one **year**.

The sun seems to rise in the **east** and set in the **west** every day, because the Earth is spinning.

3 The moon

The **moon** is the Earth's natural **satellite**, which means it goes around the Earth. One trip takes about a **month**.

The moon does not give out its own light. It **reflects** sunlight.

Half of the moon is always lit, but from the Earth we see different amounts of the lit half. That is why the moon seems to change **shape**: **new moon, crescent, half moon and full moon**.

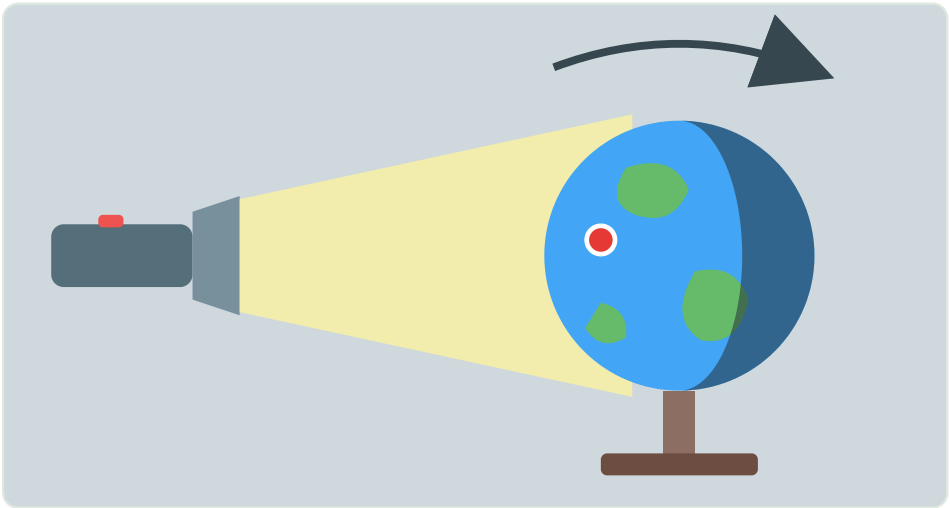
Islamic months begin with the moon. A thin crescent is looked for at the start of each new month.

There is no air or water on the moon, so nothing lives there.

Key words

English	سنڌي	اردو
solar system	شمسي نظام	نظامِ شمسی
star	تارو	ستارہ
planet	سيارو	سیارہ
satellite	سيٽلائيٽ	سیارچہ
axis	محور	محور
rotation	گردش	محوری گردش
revolution	سج جو چڪر	سالانہ گردش
east	اوپر	مشرق
west	اولهه	مغرب
crescent	هلال	ہلال
full moon	پورو چنڊ	پورا چاند

Experiment: Day and night with a torch and a ball



You need

- A torch (the sun)

Steps

1. Put a small mark on the ball. This is your home.

- A ball or an orange (the Earth)
- A small sticker or a pen mark
- A dark room

2. Turn off the room lights and ask a friend to shine the torch at the ball.
3. Slowly spin the ball.
4. Watch when the mark is in the light and when it is in the dark.

Safety: Move carefully in the dark and never shine the torch into anyone's eyes. Never look straight at the sun.

What did we learn? When the mark faces the torch it is in the light (day). As the ball spins and the mark moves to the other side, it is in the dark (night). Day and night happen because the Earth spins.

Practice questions

Circle the correct answer

1. **What is the sun?**

   (a) A star **(b)** A planet
   (c) A moon **(d)** A cloud

2. **How many planets are there in the solar system?**

   (a) 6 **(b)** 8
   (c) 10 **(d)** 12

3. **Which is the biggest planet?**

   (a) Earth **(b)** Mars
   (c) Jupiter **(d)** Mercury

4. **Which planet is closest to the sun?**

   (a) Mercury **(b)** Venus
   (c) Neptune **(d)** Saturn

5. **Why do we have day and night?**

   (a) The Earth spins on its axis **(b)** The sun switches off
   (c) The moon covers the sun **(d)** Clouds hide the sun

6. **How long does the Earth take to spin once on its axis?**

   (a) About 24 hours **(b)** One week
   (c) One month **(d)** 365 days

7. **How long does the Earth take to go once around the sun?**

(a) 24 hours **(b)** 30 days

(c) About 365 days **(d)** 10 years

8. **In which direction does the sun seem to rise?**

(a) East **(b)** West

(c) North **(d)** South

9. **What is the moon?**

(a) The Earth's natural satellite **(b)** A star

(c) A piece of the sun **(d)** A planet

Write true or false

10. **Earth is the only planet where we know there is life.** _____

11. **The moon makes its own light.** _____

12. **It is day on the side of the Earth that faces the sun.** _____

13. **There is lots of air and water on the moon.** _____

14. **Saturn has rings around it.** _____

Match the columns

15. Match each word with its meaning.

1. Rotation		(a) One trip around the sun
2. Year		(b) Something that goes around a planet
3. Satellite		(c) Earth spinning on its axis

Match the columns

16. Match each planet with a fact about it.

1. Mars		(a) The biggest planet
2. Jupiter		(b) Has beautiful rings
3. Saturn		(c) Has life
4. Earth		(d) The red planet

Worksheet: Earth and Space

Name: _____ Date: _____

A. Fill in the blanks

Word box: 24 · east · eight · satellite · star

1. The sun is a _____ .
2. There are _____ planets in the solar system.
3. The Earth spins once on its axis in about _____ hours.
4. The moon is the Earth's natural _____ .
5. The sun seems to rise in the _____ .

B. Put ✓ for true and ✗ for false

1. ☐ Jupiter is the smallest planet.
2. ☐ The Earth goes around the sun.
3. ☐ The moon reflects sunlight.
4. ☐ A year has about 100 days.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Mercury	( ) Planet with rings
2. Mars	( ) Round bright moon
3. Saturn	( ) Closest to the sun
4. Full moon	( ) Red planet

D. Answer in short

1. Write the names of the eight planets in order from the sun.

2. Why do day and night happen?

3. Why is life possible on Earth?

E. Draw and label

Draw four shapes of the moon: new moon, crescent, half moon and full moon, and label them.

FOR TEACHERS

Answer key

Chapter 10: Earth and Space

Practice questions

1. (a) A star
2. (b) 8
3. (c) Jupiter
4. (a) Mercury
5. (a) The Earth spins on its axis
6. (a) About 24 hours
7. (c) About 365 days
8. (a) East
9. (a) The Earth's natural satellite
10. True
11. False
12. True
13. False
14. True
15. 1 → (c), 2 → (a), 3 → (b)
16. 1 → (d), 2 → (a), 3 → (b), 4 → (c)

Worksheet

A. Fill in the blanks: 1. star · 2. eight · 3. 24 · 4. satellite · 5. east

B. Put ✓ for true and ✗ for false: 1. ✗ · 2. ✓ · 3. ✓ · 4. ✗

C. Match column A with column B (write the number in the brackets): (3) Planet with rings · (4) Round bright moon · (1) Closest to the sun · (2) Red planet

D. Answer in short:

1. Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.
2. The Earth spins on its axis. The side facing the sun has day and the other side has night.
3. Because it has water, air and the right amount of heat.

Picture credits

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